

DBMS LAB 3

Parth Janardan Naukudkar_ECS_II_66_D6

Aim: To design and represent an Entity Relationship (ER) diagram for a real-world database system using the draw.io tool by identifying entities, attributes, keys, relationships and cardinalities.

1. Identify all entity types and entity sets.

Researcher- Represents all researchers working in the university.

Department- Represents all departments in the university.

Research Project- Represents all research projects undertaken by the university.

Student- Represents all students working under researchers.

Publication- Represents all publications produced from research projects.

Project Assistant- Represents all assistants associated with research projects.

2. Identify strong and weak entity sets.

Strong and Weak Entity Sets

- Strong Entity Sets:
 - Researcher
 - Department
 - Research Project
 - Student
 - Publication
- Weak Entity Set:
 - Project Assistant

Project Assistant is a weak entity because it is identified only within the context of a particular Research Project.

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3. Identify the owner entity and partial key of the weak entity.

Weak Entity: Project Assistant

Owner Entity: Research Project

Partial Key: Assistant_Name

The complete identification of a Project Assistant is:

Project_ID + Assistant_Name

4. Simple attributes.

The simple attributes in the University Research Management System are:

- **Researcher:** Researcher_ID, Name, Date_of_Birth, Email
- **Department:** Department_ID, Department_Name, Location
- **Research Project:** Project_ID, Project_Title, Start_Date, End_Date, Budget
- **Student:** Student_ID, Name, Email, Date_of_Birth
- **Publication:** Publication_ID, Title, Publication_Date, DOI, Publisher
- **Project Assistant:** Assistant_Name, Role, Joining_Date

5. Identify composite attributes.

The composite attribute in the system is:

1. Researcher → Address

- Street
- City
- State
- PIN_Code

Address is a composite attribute because it can be divided into smaller component attributes.

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Name is treated as a simple attribute because the given problem statement does not specify First_Name, Middle_Name, and Last_Name.

6. Identify single-valued attributes.

Researcher

- Researcher_ID
- Name
- Date_of_Birth
- Email
- Address

Department

- Department_ID
- Department_Name
- Location

Research Project

- Project_ID
- Project_Title
- Start_Date
- End_Date
- Budget

Student

- Student_ID
- Name
- Email
- Date_of_Birth

Publication

- Publication_ID
- Title
- Publication_Date
- DOI
- Publisher

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Project Assistant

- Assistant_Name
- Role
- Joining_Date

7. Identify multi-valued attributes.

The multi-valued attributes in the system are:

1. Researcher → Phone_Number

A researcher can have multiple phone numbers.

2. Research Project → Keyword

A research project can have multiple keywords.

8. Identify stored attributes.4

Researcher

- Researcher_ID
- Name
- Date_of_Birth
- Email
- Phone_Number
- Address
 - Street
 - City
 - State
 - PIN_Code

Department

- Department_ID
- Department_Name
- Location

Research Project

- Project_ID
- Project_Title
- Start_Date
- End_Date
- Budget
- Keyword

Student

- Student_ID
- Name
- Email
- Date_of_Birth

Publication

- Publication_ID
- Title
- Publication_Date
- DOI
- Publisher

Project Assistant

- Assistant_Name
- Role
- Joining_Date

9. Identify derived attributes.

Derived Attribute – Age

Age is derived from **Date_of_Birth**.

Therefore, Age does not need to be stored separately in the database.

In the ER diagram, **Age** is represented using a **dashed oval**.

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10. Identify key attributes.

Entity	Key Attribute	Key Status
Researcher	Researcher_ID	Primary Key
Department	Department_ID	Primary Key
Research Project	Project_ID	Primary Key
Student	Student_ID	Primary Key
Publication	Publication_ID	Primary Key
Project Assistant	Assistant_Name	Partial Key
Project Assistant	Project_ID + Assistant_Name	Composite Key

11. Identify candidate keys and select appropriate primary keys.

Entity	Candidate Key	Selected Primary Key
Researcher	Researcher_ID	Researcher_ID
Department	Department_ID	Department_ID
Research Project	Project_ID	Project_ID
Student	Student_ID	Student_ID
Publication	Publication_ID	Publication_ID
Project Assistant	Project_ID + Assistant_Name	Project_ID + Assistant_Name

12. Identify at least one composite key.

The composite key is:

Project Assistant → Project_ID + Assistant_Name

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Assistant_Name alone cannot uniquely identify an assistant because the same assistant name may occur in different projects.

Therefore:

Composite Key = (Project_ID, Assistant_Name)

Another composite key occurs in the relational table created for the M:N **WORKS_ON** relationship:

(Researcher_ID, Project_ID)

13. Identify appropriate foreign keys after converting the ER model into relations.

Relation	Foreign Key	References
RESEARCHER	Department_ID	DEPARTMENT(Department_ID)
WORKS_ON	Researcher_ID	RESEARCHER(Researcher_ID)
WORKS_ON	Project_ID	RESEARCH_PROJECT(Project_ID)
STUDENT	Researcher_ID	RESEARCHER(Researcher_ID)
PUBLICATION	Project_ID	RESEARCH_PROJECT(Project_ID)
PROJECT_ASSISTANT	Project_ID	RESEARCH_PROJECT(Project_ID)
RESEARCHER_PHONE	Researcher_ID	RESEARCHER(Researcher_ID)
PROJECT_KEYWORD	Project_ID	RESEARCH_PROJECT(Project_ID)

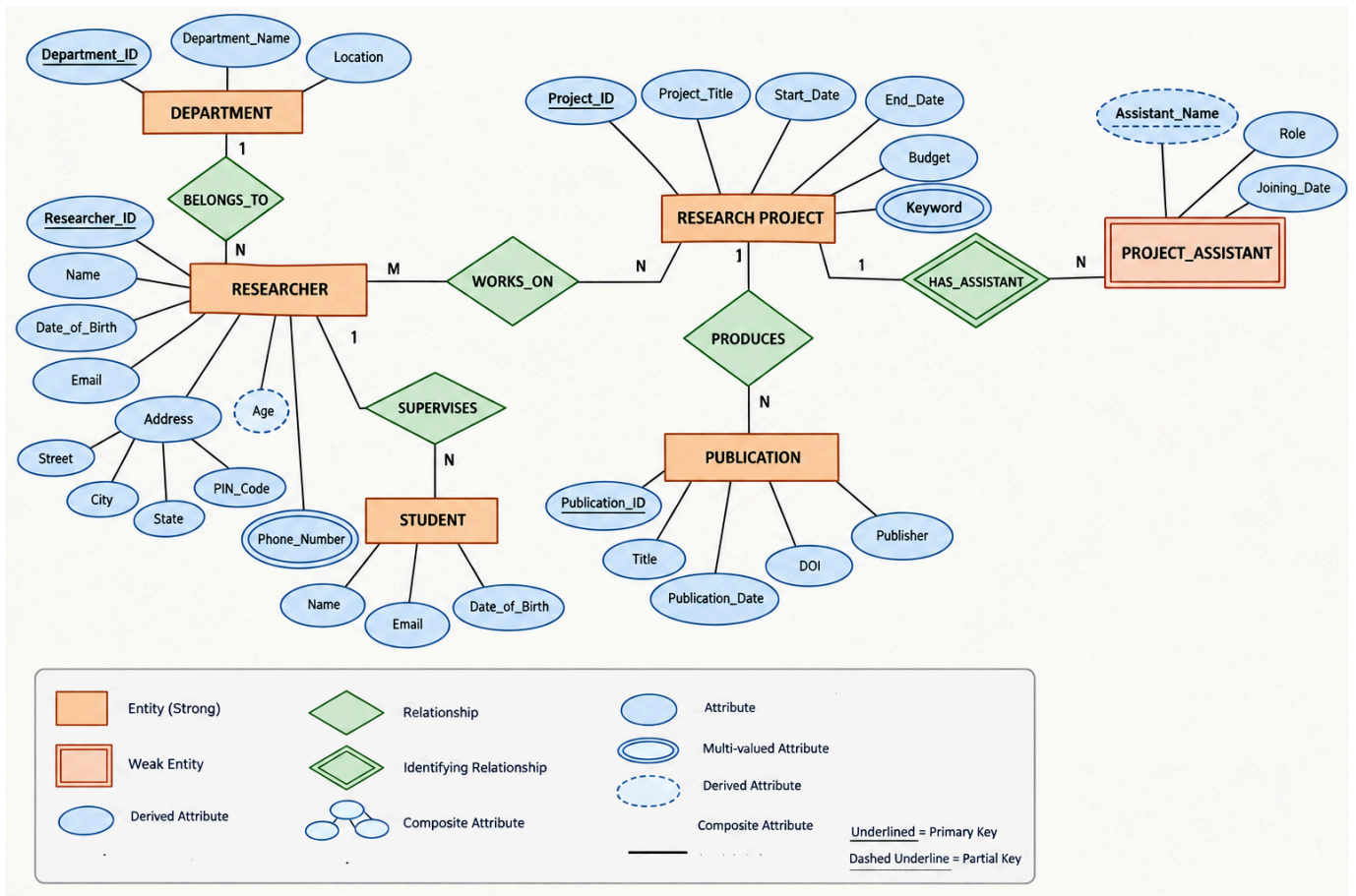
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14. Identify all major relationships between the entities.

Relationship	Entities Involved	Cardinality	Meaning
BELONGS_TO	Department ↔ Researcher	1 : N	One department has many researchers; each researcher belongs to one department.
WORKS_ON	Researcher ↔ Research Project	M : N	A researcher can work on many projects, and a project can involve many researchers.
SUPERVISES	Researcher ↔ Student	1 : N	One researcher can supervise many students; each student works under one researcher.
PRODUCES	Research Project ↔ Publication	1 : N	One project can produce many publications; each publication belongs to one project.
HAS_ASSISTANT	Research Project ↔ Project Assistant	1 : N	One project can have many assistants; each assistant belongs to one project. This is an identifying relationship because Project Assistant is a weak entity.

15. Draw the complete ER diagram using standard ER notation.



16. Prepare a table showing each attribute and its attribute type and key status.

Entity	Attribute	Attribute Type	Value	Key Status
Researcher	Researcher_ID	Simple	Single	Primary Key
Researcher	Name	Simple	Single	Non-key
Researcher	Date_of_Birth	Simple	Single	Stored
Researcher	Age	Derived	Single	Derived
Researcher	Email	Simple	Single	Non-key
Researcher	Phone_Number	Simple	Multi-valued	Non-key
Researcher	Address	Composite	Single	Non-key
Researcher	Street	Simple	Single	Non-key

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Entity	Attribute	Attribute Type	Value	Key Status
		component		
Researcher	City	Simple component	Single	Non-key
Researcher	State	Simple component	Single	Non-key
Researcher	PIN_Code	Simple component	Single	Non-key
Department	Department_ID	Simple	Single	Primary Key
Department	Department_Name	Simple	Single	Non-key
Department	Location	Simple	Single	Non-key
Research Project	Project_ID	Simple	Single	Primary Key
Research Project	Project_Title	Simple	Single	Non-key
Research Project	Start_Date	Simple	Single	Stored
Research Project	End_Date	Simple	Single	Stored
Research Project	Budget	Simple	Single	Stored
Research Project	Keyword	Simple	Multi-valued	Non-key
Student	Student_ID	Simple	Single	Primary Key
Student	Name	Simple	Single	Non-key
Student	Email	Simple	Single	Non-key
Student	Date_of_Birth	Simple	Single	Stored
Publication	Publication_ID	Simple	Single	Primary Key

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Entity	Attribute	Attribute Type	Value	Key Status
Publication	Title	Simple	Single	Non-key
Publication	Publication_Date	Simple	Single	Stored
Publication	DOI	Simple	Single	Non-key
Publication	Publisher	Simple	Single	Non-key
Project Assistant	Assistant_Name	Simple	Single	Partial Key
Project Assistant	Role	Simple	Single	Non-key
Project Assistant	Joining_Date	Simple	Single	Non-key